Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

EIN 4360 Facilities Planning and Material Handling

Module 10 – Schedule Design REVISITED

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POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

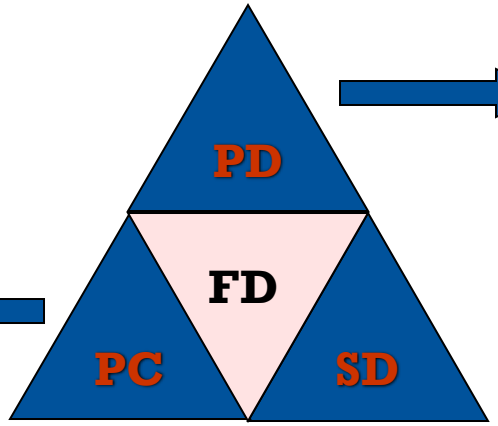
Summary

In this lecture we will revisit Chapter 2 on Schedule Design to learn about:

1. Scrap Estimates
2. Reject Allowance Problem (RAP)
3. Machine Fractions
4. Machine Assignment Problem

Relationship between Product Design (PD), Process Design (PC) & Schedule Design (SD) and Facilities Design (FD)

RECALL

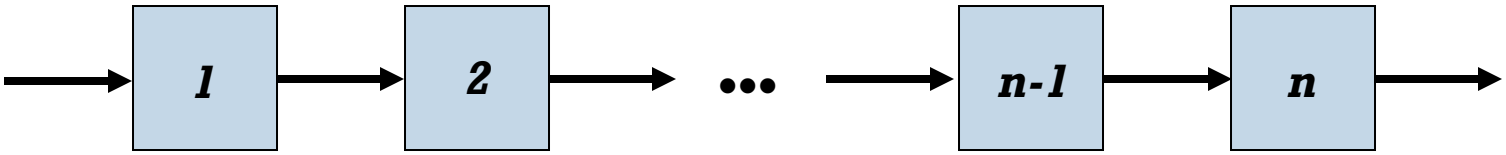


- Products to be manufactured
- Detailed design of individual parts

Manufacturing processes for each part

- Lot sizes
- Production schedule
- Machine requirements

Operational Sequence for Each Part



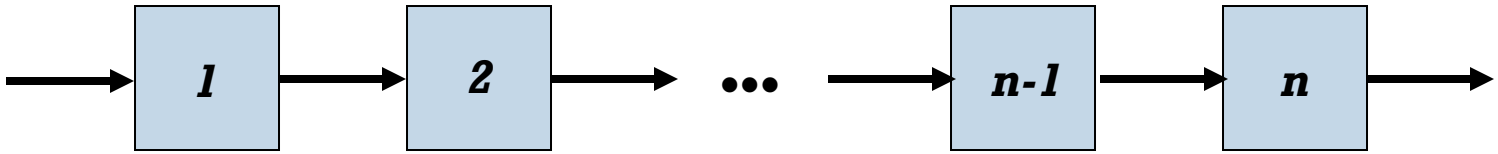
The outputs from the process design (processes, equipment & raw materials req'd for in-house production of products) are generally given in the form of a route sheet.

Schedule Design

SC = Quantities to be produced & Machine requirements

1. How much
2. When

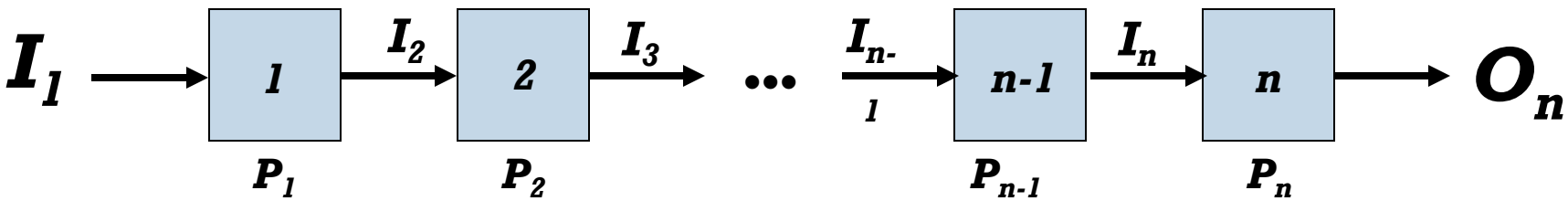
1. Number of each machine type
2. Fractional machines



P_k = percent defective for machine k

I_k = number of parts processed by machine k

O_n = number of acceptable parts after n operations

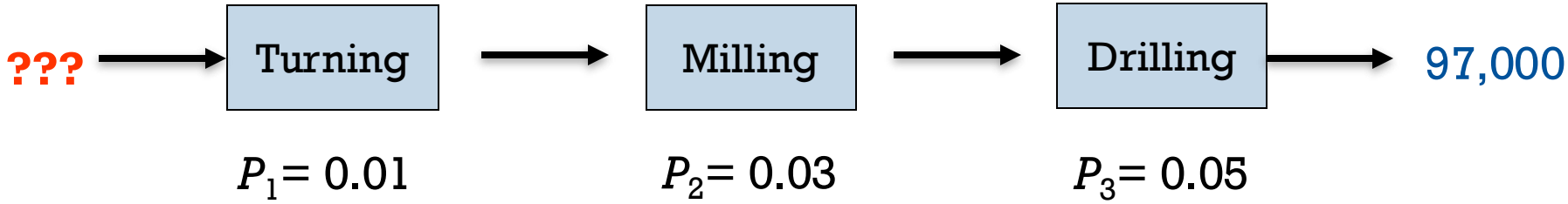


$$I_1 = \frac{O_n}{\prod_{k=1}^{n}(1-P_k)}$$

Good for production layout

Example – Scrap Estimates

- A product has a market estimate of 97,000 components and requires 3 processing steps (turning, milling, and drilling) having scrap estimates of $P_1 = 0.01$, $P_2 = 0.03$, and $P_3 = 0.05$. How many components should be produced to meet the market demand?



I=106,300

Reject Allowance Problem

- Important problem for job shops where "one-shot" jobs are received and processed
  - Typically no inventory is maintained
  - Order processed and shipped to customer
  - Too many vs. too few produced
  - Production lead time constraint – no second attempt for producing "remaining" jobs
  - Expected profit model – determine the optimum batch size
    - Balance the cost of producing too many against too few units

Formulation

X = R.V. denoting the number of good units produced

$p(x)$ = probability of producing exactly x good units

Q = production lot size

$R(Q,X)$ = revenue from producing Q units of which exactly X are good

$C(Q,X)$ = cost of producing a lot of size Q, of which exactly X are good

$P(Q,X)$ = profit resulting from producing Q units, of which exactly X are good = $R(Q,X) - C(Q,X)$

$E[R(Q)]$ = expected revenue resulting from producing Q units

$E[C(Q)]$ = expected cost resulting from producing Q units

$E[P(Q)]$ = expected profit resulting from producing Q units = $E[R(Q)] - E[C(Q)]$

Expected profit model: $$\boxed{E[P(Q)] = \sum_{x=0}^{Q} \{R(Q,x) - C(Q,x)\} p(x)}$$

Determine the value of Q that maximizes the profit model

Example

Suppose that a small foundry receives an order for 20 specially designed castings. The owner of the foundry quotes a price of $4000 each for the castings. Only one production run can be made due to the customer's need for the castings "right away"! If fewer than 18 good castings are produced, the customer will not accept any of the castings; if 18, 19, or 20 good castings are produced, the customer will purchase all good castings; if more than 20 good castings are produced, the customer will only pay for 20. The remaining castings whether good or bad, can be recycled in the foundry; hence each has a value equal to the material cost of $500. In addition to the material cost, the cost of producing a casting is estimated to be $2250. The probability mass function for the number of good castings is given below. The probabilities are based on past work performed of similar complexity. What should be the production lot size if expected profit is to be maximized?

PROBABILITY MASS FUNCTION																	
	Number of good units produced																
Lot size	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
20	0.1	0.1	0.1	0.2	0.2	0.2	0.1										
21	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1									
22		0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1								
23			0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1							
24			0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1						
25						0.1	0.1	0.1	0.2	0.2	0.2	0.1					
26						0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1				
27							0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1			
28								0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1		
29								0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1	
30											0.1	0.1	0.1	0.2	0.2	0.2	0.1

Example: Problem Formulation

Revenue Function:

$$R(Q,x)=\begin{cases}\$500Q, & x=0,\ldots,17\\ \$4000x+\$500(Q-x), & x=18,19,20\\ \$4000(20)+\$500(Q-20), & x=21,\ldots,Q\end{cases}$$

Cost Function: $C(Q,x) = \$(500 + 2250)Q = \$2750Q$

Expected Profit Formula: $E[P(Q)]=\sum_{x=0}^{Q}\{R(Q,x)p(x)\}-C(Q)$

Expected Revenue: $E[R(Q)]=\$500Q+\$3500\sum_{x=18}^{20}xp(x)+\$70{,}000\sum_{x=21}^{Q}p(x)$

Expected Profit: $E[P(Q)]=\$3500\sum_{x=18}^{20}xp(x)+\$70{,}000\sum_{x=21}^{Q}p(x)-\$2250Q$

Solution

	PROBABILITY MASS FUNCTION																				
	Number of good units produced																				
Lot size	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	3500Σxp(x)	70,000Σp(x)	Expected Profit
20	0.1	0.1	0.1	0.2	0.2	0.2	0.1											32,900		(12,100)
21	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1										32,900	7,000	(7,350)
22		0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1									33,600	14,000	(1,900)
23			0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1								26,950	28,000	3,200
24			0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1							19,950	35,000	950
25						0.1	0.1	0.1	0.2	0.2	0.2	0.1						13,650	56,000	**13,400**
26						0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1					13,650	56,000	11,150
27							0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1				7,000	63,000	9,250
28								0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1				70,000	7,000
29								0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1			70,000	4,750
30											0.1	0.1	0.1	0.2	0.2	0.2	0.1		70,000	2,500

In-class Activity

You want to determine how many cookies to bake for a catering order. The client wants 100 cookies but is willing to pay up to 110 cookies. If there are fewer than 100 cookies, they will refuse the order. Each cookie costs 50 cents to make and is sold at 1 dollar. Assume any cookies not sold could be donated for a tax write-off of 25 cents each. Use the probability mass function provided in canvas to answer the question: How many cookies should you bake?

In-class Activity

Revenue Function:

$$R(Q,x) = \begin{cases} 0.25Q & x = 0, \ldots, 99 \\ 1x + 0.25(Q - x) & x = 100, \ldots, 110 \\ 110 + 0.25(Q - 110) & x = 111, \ldots, Q \end{cases}$$

Cost Function: $C(Q,x) = \$0.50Q$

Expected Profit Formula: $E[P(Q)] = \sum_{x=0}^{Q} \{R(Q,x)p(x)\} - C(Q)$

Expected Revenue: $E[R(Q)] = 0.25Q + 0.75 \sum_{x=100}^{110} xp(x) + 82.5 \sum_{x=111}^{Q} p(x)$

Expected Profit: $E[P(Q)] = 0.75 \sum_{x=100}^{110} xp(x) + 82.5 \sum_{x=111}^{Q} p(x) - 0.25Q$

In-class Activity

	Number of good units produced															
Lot size	100	101	102	103	104	105	106	107	108	109	110	**-0.25Q**	**0.75∑xp(x)**	**82.5∑p(x)**	**Expected P**
115	0.3	0.2	0.1	0.05	0	0	0	0	0	0	0	-28.75	49.1625	0	20.4125
116	0.2	0.3	0.2	0.1	0.05	0	0	0	0	0	0	-29	64.65	0	35.65
117	0.1	0.2	0.3	0.2	0.1	0.05	0	0	0	0	0	-29.25	72.7875	0	43.5375
118	0.05	0.1	0.2	0.3	0.2	0.1	0.05	0	0	0	0	-29.5	77.25	0	47.75
119	0	0.05	0.1	0.2	0.3	0.2	0.1	0.05	0	0	0	-29.75	78	0	48.25
120	0	0	0.05	0.1	0.2	0.3	0.2	0.1	0.05	0	0	-30	78.75	0	48.75
121	0	0	0	0.05	0.1	0.2	0.3	0.2	0.1	0.05	0	-30.25	79.5	0	49.25
122	0	0	0	0	0.05	0.1	0.2	0.3	0.2	0.1	0.05	-30.5	80.25	0	49.75
123	0	0	0	0	0	0.05	0.1	0.2	0.3	0.2	0.1	-30.75	76.8375	4.125	50.2125
124	0	0	0	0	0	0	0.05	0.1	0.2	0.3	0.2	-31	69.225	12.375	50.6
125	0	0	0	0	0	0	0	0.05	0.1	0.2	0.3	-31.25	53.2125	28.875	50.8375
126	0	0	0	0	0	0	0	0	0.05	0.1	0.2	-31.5	28.725	53.625	50.85
127	0	0	0	0	0	0	0	0	0	0.05	0.1	-31.75	12.3375	70.125	50.7125
128	0	0	0	0	0	0	0	0	0	0	0.05	-32	4.125	78.375	50.5

Space Requirements

- Equipment Requirements
  - Number of machines required for the desired production rate at each processing stage
    - P_{ij} = desired production rate for product *i* on machine *j*
    - T_{ij} = production time per product *i* on machine *j*
    - H_{ij} = number of hours available for production, including an allowance for machine downtime, etc.
    - M_j = number of machines of type *j* required
    - *N* number of products
  - Then,

$$M_j = \sum_{i=1}^{n} \frac{P_{ij} T_{ij}}{H_{ij}}$$

  - Could also use probability distributions instead of point estimates

Workstation Planning

- For each machine, determine space needed
  - Equipment footprint
  - Operator access
  - Maintenance access
  - Tooling and supplies
  - Setup area
  - In-process material
  - Scrap

Equipment Fractions

The quantity of equipment required for operation is referred to as the equipment fraction.

***F* = number of machines required per shift**

***S* = standard time (minutes) per unit**

***Q*= number of units processed per shift**

***E* = performance level (as a % of std. time)**

***H* = minutes available per machine ("up time") per shift**

***R* = availability of machine, as a fraction of "up time"**

$$F = \frac{SQ}{EHR}$$

Example

A machine part has a standard machinery time of 2.8 min per part on a milling machine. During an 8-hr shift 200 units are to be produced. Of the 480 min available for production, the milling machine will be operational 80% of the time. During the time the machine is operational, parts are produced at a rate equal to 95% of the standard rate. How many milling machines are required?

Solution:

$$F = \frac{SQ}{EHR}$$

S = 2.8 min per part, Q = 200 units per shift,

H = 480 min per shift, E = 0.95, and R = 0.80

$$F = \frac{2.8(200)}{0.95(480)(0.80)} = 1.535 \text{ machines per shift}$$

EXERCISE

Find the machine fractions for the data given in the following table. **Part X** is routed from **machine B** to **machine A** and has an annual production volume of 100,000 units. **Part Y** routed from **machine A** to **machine B** and its annual production volume is 200,000 units.

	Machine A	Machine B
Part X standard time (hr)	0.10	0.01
Part Y standard time (hr)	0.06	0.06
Part X scrap estimate (%)	6	3
Part Y scrap estimate (%)	5	4
Historical efficiency (%)	95	90
Reliability factor (%)	94	97
Equipment availability (hr per yr)	2000	2000

EXERCISE

Find the machine fractions for the data given in the following table. **Part X** is routed from **machine B** to **machine A** and has an annual production volume of 100,000 units. **Part Y** routed from **machine A** to **machine B** and its annual production volume is 200,000 units.

Q_{XA} = 100,000/(1-0.06) = 106,383
Q_{XB} = 100,000/[(1-0.03)(1-0.06)] = 109,673
Q_{YA} = 200,000/[(1-0.05)(1-0.04)] = 219,298
Q_{YB} = 200,000/(1-0.04) = 208,333

F_A = (106,383*0.1)/(0.95*0.94*2000) + (219,298*0.06)/(0.95*0.94*2000)
F_A = 5.96 + 7.37 = **13.33**
F_B = (109,673*0.01)/(0.9*0.97*2000) + (208,333*0.06)/(0.9*0.97*2000)
F_B = 0.63 + 7.16 = **7.79**

In-class Activity

A part requires three processing steps on two machines in the sequence A-B-A. The demand for this part is 10,000 units per week. The company operates six days per week, eight hours per day. Given the following performance data, find the number of each machine needed to meet the demand.

Operation	Machine	Standard Time	Efficiency	Reliability	Defects
1	A	5 min	108%	98%	3%
2	B	3 min	95%	95%	5%
3	A	3 min	90%	95%	5%

In-class Activity

Q_{A1} = 10,000/(0.95*0.95*0.97) = 11,423
Q_B = 10,000/(0.95*0.95) = 11,080
Q_{A2} = 10,000/0.95 = 10,526

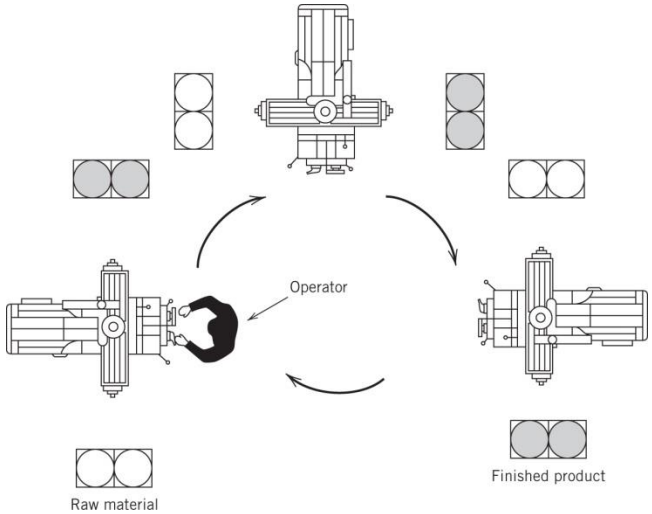
H = (8 hours/day)*(60 mins/hour)*(6 days/week) = 2,880 mins

F_A = 11,423*5/(2,880*1.08*.98) + 10,526*3/(2,880*0.9*0.95) = 31.56
F_A = 32 machines
F_B = 11,080*3/(2,880*0.95*0.95) = 12.79
F_B = 13 machines

Machine Assignment Problem

Also called **Multiple Activity Chart (MAC)** or **Human-Machine Chart**

- Consider assignment of operators to semiautomatic production equipment
- Deterministic approach
- One operator attending to more than one machine
- Balancing the load to minimize machine/operator idle time
- MAC is useful in analyzing activities during "transient" and "steady-state" conditions



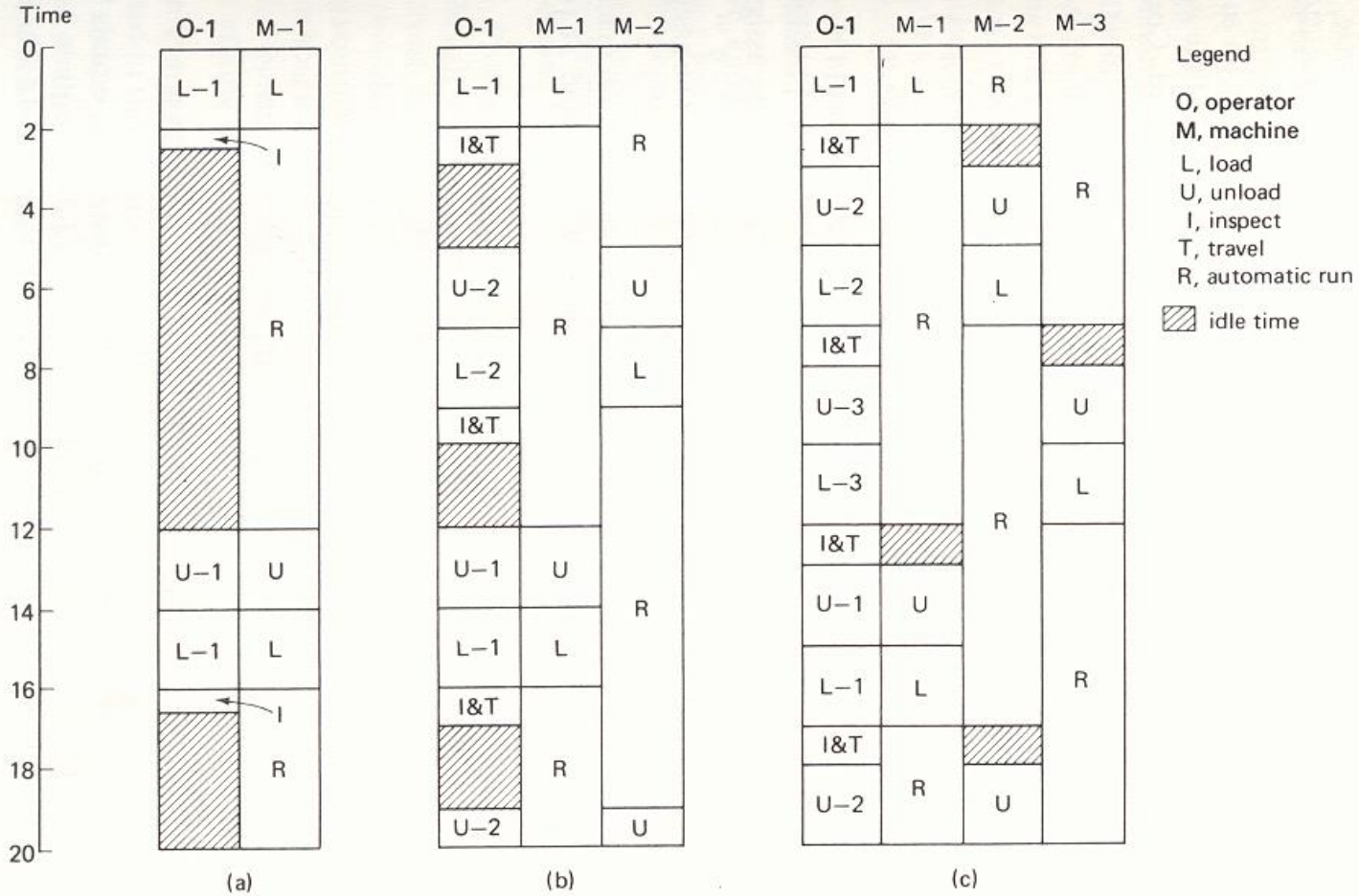


Figure 2.54 Multiple activity chart analysis of multimachine assignment.

Machine Assignment Problem

Deterministic model to for the optimum number of machines to assign to an individual operator

Mathematical formulation

a = concurrent activity time (e.g. loading/unloading a machine)
b = independent operator activity time (e.g. walking, inspecting, packing)
t = independent machine activity time (e.g. automatic machine time)
n' = ideal number of identical machines to assign an operator
m = number of machines assigned to an operator
T_c = cycle time
I_o = idle operator time during a repeating cycle
I_m = idle time for each machine during a cycle

Each machine cycle requires $a + t$ min
Operator requires $a + b$ min
For an ideal assignment, $n' = (a+t)/(a+b)$

$$T_c = a + t \quad \text{for } m \le n'$$
$$= m\,(a+b) \quad \text{for } m > n'$$

$$I_m = 0 \quad \text{for } m \le n'$$
$$= T_c - (a+t) \quad \text{for } m > n'$$

$$I_o = T_c - m(a+b) \quad \text{for } m \le n'$$
$$= 0 \quad \text{for } m > n'$$

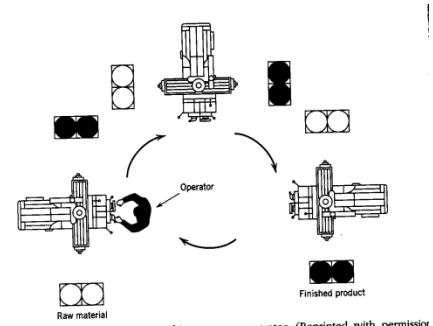


Figure 2.18 Assignment of three machines to one operator. (Reprinted with permission from [23].)

Cost Assessment for MAC

C_o = cost per operator hour

C_m = cost per machine hour

$\varepsilon = C_o/C_m$

$TC(m)$ = cost *per unit produced* based on an assignment of m machines per operator

$\Phi = TC(n)/TC(n+1)$

$TC(m) = (C_o+mC_m)(a+t)/m$ for $m \le n'$

$= (C_o+mC_m)(a+b)$ for $m > n'$

$\Phi = (C_o+nC_m)(a+t)/n[C_o+(n+1)C_m](a+b)$

$= (\varepsilon + n)\, n'/(\varepsilon + n + 1)n$

If $\Phi < 1$, then $TC(n) < TC(n+1)$, so assign n machines

If $\Phi > 1$, assign $n + 1$ machines

What happens when $\Phi = 1$?

MAC Example 1

Concurrent activity time (a) = 4 minutes
Independent machine activity time (t) = 10 minutes
Independent operator activity time (b) = 1 minute
Cost per operator hour (C_o) = \$3
Cost per machine hour (C_m) = \$10

A. *Determine ideal assignment (n').*
B. *How many machines should be assigned per operator to minimize costs?*

MAC Example 1

Concurrent activity time (a) = 4 minutes
Independent machine activity time (t) = 10 minutes
Independent operator activity time (b) = 1 minute
Cost per operator hour (C_o) = \$3
Cost per machine hour (C_m) = \$10

A.
n' = (a+t)/(a+b) = (4+10)/(1+4) = 2.8

B.
One way to do it:
Φ = TC(n)/TC(n+1) = TC(2)/TC(3)
TC(2) = (3+2*10)(4+10)/2 = 161
TC(3) = (3+3*10)(4+1) = 165
Φ = 161/165 < 1, therefore just 2 machines per operator is most cost effective.

In-class Activity

Semiautomatic mixers are used in a paint plant. It takes 6 minutes for an operator to load the pigments and paint base into a mixer. Then, mixers run automatically and automatically dispense paint into drums. Mixing and unloading take 30 minutes to complete. Mixers are cleaned *between batches*; it takes 4 minutes for the operator to clean each mixer (while the mixers are shut off). *During batches*, an operator places empty drums in the magazine to position them for filling; it takes 6 minutes to load the drums into the magazine.

a. What is the maximum number of mixers that can be assigned to an operator without creating idle time for the mixers?
b. If C_o=\$12/hr and C_m=\$25/hr, what assignment of mixers to operators will minimize the cost per batch produced?

In-class Activity

Solution

a) a=6+4=10mins, b=6mins, t=30mins
n'=(10+30)/(10+6)=2.5, so we pick 2 to avoid creating idle time for mixers

b) C_o=\$12/hr, C_m=\$25/hr, C_o/C_m=0.48
Φ = [(2.5)(2+0.48)]/[(3+0.48)(2)] = 0.89<1
hence, 2 mixers should be assigned to the operator